

DRY ETCHING

With dry etching, the material is not eroded by the compounds dissolved in a liquid, but by the atoms or molecules of a, at least, partially ionised gas.

This chapter describes the basic physical and chemical processes of this etching process and sets out the specific requirements of the previous photoresist processing to define of the desired resist mask.

Basic Etching Mechanisms and Parameters

Physical and Chemical Processes in Dry Etching

The erosion of the material to be etched can either be carried out physically by the kinetic energy of the particles (neutral or ionized atoms or molecules) from the gas, or through chemical reactions between the material and the gas.

The physical mechanism can be regarded as a partial-elastic collision of ions with the atoms of the medium to be etched. Chemical reactions play no or only a subordinate role, which is why etching is hardly materially selective. Since the ions from the gas are usually accelerated by electric fields perpendicular to the substrate, the material removal in this preferred direction is anisotropic.

If the 'chemical' mechanism dominates, etching occurs via the strong material selective formation of volatile compounds by radicals in the plasma which – towards high plasma pressure – hit the surface more and more isotropically. Compared to the physical erosion, the chemical erosion allows a significantly higher etching rate.

Dry Etching Technologies

Plasma etching is dominated by chemical erosion. In this way Si or SiO₂ is etched usually with chlorinated and fluorinated hydrocarbons isotropic and very material selectively.

With *sputter etching (ion milling)*, the material is eroded physically by inert gas ions accelerated on the substrate.

RIE (Reactive Ion Etching) represents a combination of physical and chemical erosion: Here, chemically reactive radical is formed in plasma and accelerated towards the substrate.

Criteria with Dry Etching

The *etching rate* is defined by the eroded thickness of the material to be etched per time.

The *selectivity* is the ratio of the etching rate of two materials that are simultaneously exposed to the etching. This can be, for example, the photoresist structures of the etching mask, as well as the material to be etched, but also two vertically stacked materials that are to be etched in succession.

The *anisotropy* describes the ratio of etch rates perpendicular and parallel to the substrate surface. The more isotropic the etching process is, the stronger the etching mask is underetched during the etching process.

The *uniformity* describes the homogeneity of the etching rate over the entire substrate.

Plasma Etching of Si and SiO₂

Typical Etching Gases

Typical etching gases for the etching of SiO₂ are fluorinated hydrocarbons (C_xF_yH_z) such as tetrafluoromethane (CF₄)₂. The basic reactions in plasma and on the SiO₂ to be etched are:

(1) Formation of F-radicals through electron impacts: $e + CF_4 \rightarrow CF_3 + F + e^-$

(2) Formation of volatile Si compounds: $SiO_2 + 4 F \rightarrow SiF_4 + O_2$

Typical etching gases for etching of Si are chlorinated and fluorinated carbon compounds (C_xF_yCL_z), again using the example of tetrafluoromethane:

(1) Formation of F-radicals through electron impacts: $e + CF_4 \rightarrow CF_3 + F + e^-$

(2) Formation of volatile Si compounds: $Si + 4 F \rightarrow SiF_4$

Adjustment of the Required Etching Rates Ratio Si : SiO₂

The addition of oxygen increases via the reaction $\text{CF}_3 + \text{O} \rightarrow \text{COF}_2 + \text{F}$ the concentration of fluorine radicals in the plasma with the following consequences:

- Via the reaction $\text{Si} + 4 \text{F} \rightarrow \text{SiF}_4$ increases the etching rate of silicon to a maximum with a percentage of approx. 12% O₂ in CF₄.
- Via the reaction $\text{SiO}_2 + 4 \text{F} \rightarrow \text{SiF}_4 + \text{O}_2$, the etching rate of SiO₂ increases with a maximum of approximately 20% O₂ in CF₄.
- Via O₂ combustion, an existing resist mask is eroded more strongly

The addition of hydrogen to the process gas

- reduces the concentration of fluorine radicals in the plasma via the reaction $\text{H} + \text{F} \rightarrow \text{HF}$ and lowers the etching rate, however, for Si more than for SiO₂.
- leads to the chemically very inert fluorinated polymer deposition on Si surfaces via the reaction $\text{CF}_4 + \text{H} + \text{Si} \rightarrow \text{CH}_x\text{F}_y$ thus stopping the etching of silicon.

Deep Reactive Ion Etching: The "Bosch Process"

The so-called *Bosch Process* lends itself to the dry chemical etching of structures with steep sidewalls and a very high aspect ratio.

Alternating anisotropic Si etching and the formation of a fluorinated polymer layer (which is inert against the plasma) on the etched sidewalls as well as the sidewalls of the resist structures allows aspect ratios > 50, Si etching rates > 10 µm/min, etching rate ratios > 450 (Si : SiO₂) and > 150 (Si : photoresist).

Plasma Etching of Certain Metals

Aluminium

Aluminium can be etched using gases such as hydrogen bromide (HBr) or chlorine-containing gases such as under the formation of sufficiently volatile compounds of aluminium bromide (AlBr₃) or aluminium chloride (AlCl₃).

Tungsten

Tungsten is etched with fluorine-containing gases with the formation of volatile tungsten hexafluoride (WF₆), the densest known gas under standard conditions.

Titanium

Due to the very low vapour pressures of titanium chloride (TiCl₃) and titanium fluoride (TiF₃), pure plasma processes with accordingly halogenated process gases are not suitable for the dry etching of titanium which is why argon is usually added to increase the erosion with sputter etching.

Copper, Silver, and Gold

These metals do not form sufficiently volatile halides for adequately high plasma etching rates at temperatures below 150°C. With the help of hydrogenous process gases however (unstable) hydrides of the metals can form, which via ion or photon-assisted processes can be desorbed from the surface at etching rates of a few nm/min.

Photoresist Processing Requirements

Vertical Resist Sidewalls

For the steepest possible resist profiles, a high-contrast, photoresist, as well as process parameters optimised for high contrast are required, i.e.

- depending on the desired resist film thickness and required thermal stability of the AZ® 701 MiR for resist film thicknesses below 1 µm, the AZ® ECI 3000 series for 1 - 3 µm resist film thickness, or AZ® 9260 for even thicker layers,
- the reduction of the dark erosion and maintenance of a possible high development rate of positive

resists via, among other things, optimised softbake parameters, and

- the use of a highly selective, i.e. optimally diluted developer such as the MIF developers AZ® 326 and AZ® 726 or the AZ® 400K or AZ® 351B in a sufficiently high dilution.

Resist Lenses

If ellipsoid resist structures in the substrate are to be transferred, a normally processed resist profile with a rectangular cross-section is usually softened by heating via the softening temperature of the resist. For this process, all positive resists are suitable. The series AZ® 1500, AZ® 4500 and 9200 have a relatively low softening temperature of approx. 100 - 110 °C.

Removal of the Photoresist Mask after Etching

All standard strippers are generally suitable to remove the resist mask after dry etching. In case of increased temperatures during dry etching, possibly supported by the deep UV background radiation from the plasma, the resist structures can cross-link near their surface. If the removability of the resist mask suffers after the etching process, the measures listed in the following section can be applied against excessive heating. Ultrasonic treatment during stripping also supports removal of the resist structures.

For highly cross-linked positive resists which can not be removed with standard removers, the high-performance stripper TechniStrip® P1316 is recommended for positive resists or the TechniStrip® NI555 for many Novolak based negative resists such as the AZ® nLOF 2000 series.

Measures Against the Thermal Softening of Resist Structures

Heat development during etching can soften the edges of the used photoresist mask which is transferred to the substrate during dry etching. Possible remedial measures are

- an optimised heat coupling of the substrate to its holder (e.g. some drops of turbo pump oil for proper heat transfer from strained, curved substrates)
- a sufficiently high heat buffer (massive substrate holder construction) or
- heat removal (e.g. black anodised aluminium as rear infrared radiator) from the substrate holder
- a reduced deposition rate and/or a multi-stage deposition with cooling interval(s) in between or
- a thermally more stable photoresist like the AZ® 701 MiR or the AZ® ECI 3000 series
- a sufficient softbake to minimise the residual solvent content.

Measures Against Bubble Formation in the Resist Layer during Dry Etching

Appearance

Sometimes, bubbles in the resist or even a foam-like resist appearance is observed after dry-etching. In most cases, nitrogen or evaporating solvent or water is the reason for this behaviour.

Evaporation of Residual Solvents

Another possible source of vapour bubbles is water which has penetrated during development in the resist film and can be evaporated after development with another baking. In this case, a baking step after development at approx. 80 - 100°C (always below the resist softening point!) helps reduce the water concentration and a thermal deformation of the resist structures.

Evaporation of Water

An insufficient softbake (too short/too cool) may cause the evaporation of the remaining solvent from the resist forming bubbles.

Nitrogen Formation

The developed resist structures of DNQ-based positive resists are still photo-active and can be exposed by the short-wave thermal or recombination radiation from plasma forming larger amounts of nitrogen. To ensure all photoinitiator has been converted and all nitrogen has been released from the resist film before the dry etch process, we recommend a flood exposure without mask, followed by a delay to out-

gas the nitrogen formed before the substrate is introduced to the dry etching.

Image reversal resists in image reversal mode are no longer photoactive after development and negative resists don't release nitrogen during exposure, so these resists are not affected by this problem.